

ILR-SMO: Iterative Latent Refinement for Robust Spatial Multi-Omics Integration

Author Name

Affiliation

email@example.com

A Theoretical Analysis of Iterative Latent Refinement

We analyze the stability of Iterative Latent Refinement (ILR) by modeling its update dynamics on a modality-conditioned objective.

A.1 Notation and Assumptions

We focus on the optimization behavior of the shared latent variable $L \in \mathbb{R}^d$, while treating all other network parameters as fixed or slowly varying. To avoid over-specifying architectural details, we model ILR as implicit gradient descent on a modality-conditioned objective. We consider the composite objective

$$F(L) = \sum_{m=1}^M f_m(L) + r(L), \quad (1)$$

where $f_m(L)$ denotes the modality-conditioned component objective and $r(L)$ aggregates regularization terms. We define

$$\begin{aligned} g_m(L) &= \nabla f_m(L), \\ g_r(L) &= \nabla r(L), \\ \nabla F(L) &= \sum_{m=1}^M g_m(L) + g_r(L). \end{aligned} \quad (2)$$

Assumption 1 (Smoothness). Each $f_m(L)$ is β_m -smooth:

$$\|g_m(L') - g_m(L)\| \leq \beta_m \|L' - L\|, \quad \forall L, L'. \quad (3)$$

The regularization term $r(L)$ is also β_r -smooth:

$$\|g_r(L') - g_r(L)\| \leq \beta_r \|L' - L\|, \quad \forall L, L'. \quad (4)$$

We denote $\bar{\beta} = \max(\max_m \beta_m, \beta_r)$.

Assumption 2 (Bounded gradients). There exist constants G_m and G_r such that

$$\|g_m(L)\| \leq G_m, \quad \|g_r(L)\| \leq G_r, \quad \forall L. \quad (5)$$

These assumptions are standard in non-convex optimization and suffice for analyzing the local behavior of ILR.

A.2 First-order Equivalence of Sequential Latent Updates

We compare a single-step plain joint update that use all modalities simultaneously with one full cycle of Sequential Latent Updates (SLU).

Single-step update. A single-step joint update is defined as

$$L_{\text{joint}}^+ = L - \eta \left(\sum_{m=1}^M g_m(L) + g_r(L) \right), \quad (6)$$

with update increment

$$\Delta_{\text{joint}} = L_{\text{joint}}^+ - L = -\eta \left(\sum_{m=1}^M g_m(L) + g_r(L) \right). \quad (7)$$

Sequential latent updates. Let $\sigma(1), \dots, \sigma(M)$ be a fixed permutation of modality indices. Define $L_0 = L$, and for $t = 1, \dots, M$,

$$L_t = L_{t-1} - \eta g_{\sigma(t)}(L_{t-1}). \quad (8)$$

After processing all modalities, a regularization step is applied:

$$L_{\text{SLU}}^+ = L_M - \eta g_r(L_M). \quad (9)$$

The corresponding update increment is

$$\Delta_{\text{SLU}} = L_{\text{SLU}}^+ - L. \quad (10)$$

Proposition 1 (First-order equivalence). Under Assumptions 1–2, the SLU update satisfies

$$\Delta_{\text{SLU}} = \Delta_{\text{joint}} + O(\eta^2), \quad (11)$$

and more explicitly,

$$\|\Delta_{\text{SLU}} - \Delta_{\text{joint}}\| \leq \bar{\beta} \eta^2 (M-1) \sum_{m=1}^M G_m + \beta_r \eta^2 \left(\sum_{m=1}^M G_m \right). \quad (12)$$

This bound indicates that the discrepancy between SLU and joint update is second-order in the step size η .

Proof. Define the joint update

$$\begin{aligned} L_{\text{joint}}^+ &= L - \eta \left(\sum_{m=1}^M g_m(L) + g_r(L) \right), \\ \Delta_{\text{joint}} &= L_{\text{joint}}^+ - L. \end{aligned} \quad (13)$$

For SLU, let $L_0 = L$ and for $t = 1, \dots, M$,

$$L_t = L_{t-1} - \eta g_{\sigma(t)}(L_{t-1}), \quad (14)$$

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and apply the regularization step

$$L_{\text{SLU}}^+ = L_M - \eta g_r(L_M), \quad \Delta_{\text{SLU}} = L_{\text{SLU}}^+ - L. \quad (15)$$

We first compare L_M with the one-shot modality update $L - \eta \sum_m g_m(L)$. For each t , define

$$e_t := g_{\sigma(t)}(L_{t-1}) - g_{\sigma(t)}(L). \quad (16)$$

By smoothness,

$$\|e_t\| \leq \bar{\beta} \|L_{t-1} - L\|. \quad (17)$$

Moreover,

$$L_{t-1} - L = -\eta \sum_{j=1}^{t-1} g_{\sigma(j)}(L_{j-1}), \quad (18)$$

and by bounded gradients,

$$\|L_{t-1} - L\| \leq \eta \sum_{j=1}^{t-1} \|g_{\sigma(j)}(L_{j-1})\| \leq \eta \sum_{j=1}^{t-1} G_{\sigma(j)}. \quad (19)$$

Hence,

$$\|e_t\| \leq \bar{\beta} \eta \sum_{j=1}^{t-1} G_{\sigma(j)}. \quad (20)$$

Substituting into

$$L_M = L - \eta \sum_{t=1}^M g_{\sigma(t)}(L_{t-1}) \quad (21)$$

yields

$$L_M = L - \eta \sum_{m=1}^M g_m(L) - \eta \sum_{t=1}^M e_t. \quad (22)$$

Therefore,

$$\begin{aligned} \left\| L_M - \left(L - \eta \sum_{m=1}^M g_m(L) \right) \right\| &\leq \eta \sum_{t=1}^M \|e_t\| \\ &\leq \bar{\beta} \eta^2 (M-1) \sum_{m=1}^M G_m. \end{aligned} \quad (23)$$

Now compare the full SLU update with the joint update:

$$\begin{aligned} \Delta_{\text{SLU}} - \Delta_{\text{joint}} &= \left(-\eta \sum_{t=1}^M g_{\sigma(t)}(L_{t-1}) - \eta g_r(L_M) \right) \\ &\quad - \left(-\eta \sum_{m=1}^M g_m(L) - \eta g_r(L) \right) \\ &= -\eta \sum_{t=1}^M e_t - \eta (g_r(L_M) - g_r(L)). \end{aligned} \quad (24)$$

Taking norms and using smoothness of r gives

$$\begin{aligned} \|\Delta_{\text{SLU}} - \Delta_{\text{joint}}\| &\leq \eta \sum_{t=1}^M \|e_t\| + \eta \|g_r(L_M) - g_r(L)\| \\ &\leq \eta \sum_{t=1}^M \|e_t\| + \eta \beta_r \|L_M - L\|. \end{aligned} \quad (25)$$

Finally, by bounded gradients,

$$\|L_M - L\| \leq \eta \sum_{t=1}^M \|g_{\sigma(t)}(L_{t-1})\| \leq \eta \sum_{m=1}^M G_m. \quad (26)$$

Combining the bounds completes the proof. $\square$

A.3 Robustness to Noisy Modalities

Assume that for some modality k , the gradient is perturbed as

$$\tilde{g}_k(L) = g_k(L) + \epsilon(L), \quad \|\epsilon(L)\| \leq E. \quad (27)$$

Let L_t and $\tilde{L}_t$ denote the intermediate SLU states under clean and perturbed gradients, respectively, and define $\delta_t = \tilde{L}_t - L_t$.

Proposition 2 (Bounded noise influence). Under Assumption 1, after one SLU cycle followed by the regularization step,

$$\|\tilde{L}_{\text{SLU}}^+ - L_{\text{SLU}}^+\| \leq \eta E e^{\eta \bar{\beta} M} (1 + \eta \beta_r) \leq \eta E e^{\eta(\bar{\beta} M + \beta_r)}. \quad (28)$$

The exponential bound arises from a standard relaxation of the product term and is used here for analytical clarity.

Proof. Let t_k be the index such that $\sigma(t_k) = k$. For all $t < t_k$, the updates are identical, hence $\delta_t = 0$ and in particular $\tilde{L}_{t_k-1} = L_{t_k-1}$. At the noisy step,

$$\begin{aligned} \tilde{L}_{t_k} &= \tilde{L}_{t_k-1} - \eta \tilde{g}_k(\tilde{L}_{t_k-1}) \\ &= L_{t_k-1} - \eta (g_k(L_{t_k-1}) + \epsilon(L_{t_k-1})), \\ L_{t_k} &= L_{t_k-1} - \eta g_k(L_{t_k-1}), \end{aligned} \quad (29)$$

thus

$$\delta_{t_k} = \tilde{L}_{t_k} - L_{t_k} = -\eta \epsilon(L_{t_k-1}), \quad (30)$$

and therefore $\|\delta_{t_k}\| \leq \eta E$.

For any subsequent clean modality step $t > t_k$ (i.e., $t \neq t_k$),

$$\delta_t = \tilde{L}_t - L_t \quad (31)$$

$$= (\tilde{L}_{t-1} - \eta g_{\sigma(t)}(\tilde{L}_{t-1})) - (L_{t-1} - \eta g_{\sigma(t)}(L_{t-1})) \quad (32)$$

$$= \delta_{t-1} - \eta (g_{\sigma(t)}(\tilde{L}_{t-1}) - g_{\sigma(t)}(L_{t-1})). \quad (33)$$

By Assumption 1 and $\bar{\beta} = \max(\max_m \beta_m, \beta_r)$,

$$\|\delta_t\| \leq (1 + \eta \bar{\beta}) \|\delta_{t-1}\|. \quad (34)$$

Unrolling over at most M modality steps yields

$$\|\delta_M\| \leq (1 + \eta \bar{\beta})^M \|\delta_{t_k}\| \leq (1 + \eta \bar{\beta})^M \eta E \leq \eta E e^{\eta \bar{\beta} M}. \quad (35)$$

Now consider the final regularization step:

$$L_{\text{SLU}}^+ = L_M - \eta g_r(L_M), \quad \tilde{L}_{\text{SLU}}^+ = \tilde{L}_M - \eta g_r(\tilde{L}_M). \quad (36)$$

Then

$$\|\tilde{L}_{\text{SLU}}^+ - L_{\text{SLU}}^+\| = \|\delta_M - \eta (g_r(\tilde{L}_M) - g_r(L_M))\| \quad (37)$$

$$\leq \|\delta_M\| + \eta \|g_r(\tilde{L}_M) - g_r(L_M)\|. \quad (38)$$

By smoothness of $r(L)$,

$$\|g_r(\tilde{L}_M) - g_r(L_M)\| \leq \beta_r \|\tilde{L}_M - L_M\| = \beta_r \|\delta_M\|. \quad (39)$$

Hence

$$\begin{aligned} \|\tilde{L}_{\text{SLU}}^+ - L_{\text{SLU}}^+\| &\leq (1 + \eta\beta_r) \|\delta_M\| \\ &\leq \eta E e^{\eta\bar{\beta}M} (1 + \eta\beta_r) \leq \eta E e^{\eta(\bar{\beta}M + \beta_r)}. \end{aligned} \quad (40)$$

This completes the proof. $\square$

A.4 Gradient Conflict Analysis

We analyze gradient conflict at the level of modality-conditioned components, focusing on the interaction among $\{g_m(L)\}_{m=1}^M$, while treating the regularization gradient $g_r(L)$ as an auxiliary stabilizing term. This analysis aims to characterize the optimization behavior of single-step joint latent update under heterogeneous modalities.

Consider the joint update direction induced by aggregating modality-specific gradients,

$$g_{\text{sum}}(L) = \sum_{m=1}^M g_m(L), \quad (41)$$

where $g_m(L) = \nabla f_m(L)$ denotes the gradient contributed by the m -th modality. The squared norm of the aggregated gradient admits the following decomposition:

$$\|g_{\text{sum}}(L)\|^2 = \sum_{m=1}^M \|g_m(L)\|^2 + 2 \sum_{i < j} g_i(L)^\top g_j(L). \quad (42)$$

The cross terms $g_i(L)^\top g_j(L)$ quantify the degree of alignment between gradients from different modalities. When multiple modality pairs exhibit negative inner products, i.e., $g_i(L)^\top g_j(L) < 0$, the aggregated gradient norm can be significantly reduced due to mutual cancellation, even if individual modality gradients have large magnitudes.

This phenomenon implies that single-step joint update may suffer from unstable or inefficient optimization trajectories, as the effective update direction becomes sensitive to gradient conflicts across modalities. Such conflicts are common in spatial multi-omics settings, where modalities often differ in signal quality, noise level, and semantic focus.

In contrast, Sequential Latent Updates avoids directly summing modality gradients into a single update. By applying modality-conditioned updates incrementally to the shared latent representation, ILR mitigates destructive interference among conflicting gradients and yields empirically more stable optimization behavior under heterogeneous and noisy modalities.

B Datasets

B.1 Mouse Brain Dataset

The mouse brain dataset is generated from a postnatal day 22 (P22) mouse brain tissue section and profiled using a spatial ATAC–RNA sequencing protocol, which simultaneously measures transcriptomic expression and chromatin accessibility while preserving spatial organization. The dataset contains 9,215 spatial spots, with 22,914 genes in the RNA

modality and 121,068 chromatin accessibility peaks in the ATAC modality, representing a high-dimensional and heterogeneous spatial multi-omics setting. The original dataset was released by the authors of the spatial ATAC–RNA-seq study [Zhang *et al.*, 2023]. In this work, we adopt the pre-processed version provided by COSMOS [Zhou *et al.*, 2025], which applies quality control and feature filtering to improve data reliability and downstream analysis. Notably, this pre-processed dataset differs from the raw sequencing output in both spot count and feature composition. As a result, MultiVI [Ashuach *et al.*, 2023], which requires access to raw count matrices and is designed for end-to-end modeling of original measurements, cannot be directly applied under this setting and is therefore excluded from comparisons on this dataset in the main quantitative evaluations.

For spatial domain identification, we use manual annotations derived from the annotated reference of the mouse brain coronal section in the Allen Mouse Brain Atlas as ground-truth labels [Zhou *et al.*, 2025]. These annotations provide biologically interpretable region-level domain assignments and are widely used as reference standards in spatial transcriptomics and multi-omics analyses. Throughout the paper, we adopt the following abbreviations for brain regions: cp (caudoputamen), ctx (cerebral cortex), aca (anterior cingulate area), acb (nucleus accumbens), vl (lateral ventricle), ccg (genu of corpus callosum), aco (anterior commissure, olfactory limb), and ls (lateral septal nucleus).

B.2 Human Lymph Node Dataset

The human lymph node dataset is derived from a human lymph node tissue section and generated using the 10x Genomics Visium platform. This dataset provides spatially resolved measurements of gene expression, antibody-derived tags (ADT) for protein abundance, and a corresponding H&E-stained histological image, enabling multi-modal analysis that integrates molecular profiles with tissue morphology. The dataset contains 3,484 spatial spots, including 18,085 genes in the transcriptomic modality and 31 protein markers in the ADT modality. Both the dataset and the associated ground-truth spatial domain annotations are provided by the authors of SpatialGlue [Long *et al.*, 2024], and are widely used as a benchmark for evaluating spatial multi-omics integration methods. In the main quantitative experiments reported in the paper, we focus on the RNA and ADT modalities, excluding histological images to ensure fair comparison with baselines that do not support image inputs. The image modality is incorporated separately in the foundation model extension experiments.

For data preprocessing, we follow modality-specific pipelines consistent with common practices in spatial transcriptomics analysis. For the RNA modality, genes expressed in fewer than 10 spots are first filtered out, followed by selection of the top 3,000 highly variable genes (HVGs). The expression matrix is then normalized by total counts, log-transformed, scaled, and restricted to the HVG subset. For the ADT modality, protein counts are normalized using cell-wise centered log-ratio (CLR) transformation, followed by scaling. For experiments involving histological information, the H&E image is aligned to spatial coordinates provided by the Vi-

sium platform. We extract dual-scale image patches (16×16 and 32×32) centered at each spatial spot and encode them using the UNI2 histopathology foundation model [Chen *et al.*, 2024], resulting in a 3,072-dimensional image embedding per spot. These embeddings are used as image modality inputs in the corresponding multimodal fusion experiments.

B.3 Human Hippocampus Dataset

The human hippocampus dataset is a spatial ATAC–RNA sequencing dataset generated from an adult human brain hippocampus, where chromatin accessibility and gene expression are simultaneously measured at aligned spatial pixels. This dataset captures fine-grained spatial variation across hippocampal layers and white matter regions, providing a challenging benchmark for spatial multi-omics integration due to the strong structural organization of the tissue. The original spatial ATAC–RNA-seq data were released by the authors of the corresponding experimental study [Zhang *et al.*, 2023]. In this work, we adopt the processed dataset and manual annotations provided by MultiGATE [Miao *et al.*, 2025], which curated biologically meaningful spatial domain labels by annotating hippocampal layers and white matter based on established anatomical references, prior studies, and marker gene expression. These annotations serve as the ground truth for evaluating spatial domain identification performance. The dataset contains 2,500 spatial spots, with RNA and ATAC modalities sharing the same spatial locations. The transcriptomic modality includes 7,666 genes, while the chromatin accessibility modality contains 28,270 features, resulting in substantial modality heterogeneity in both dimensionality and signal characteristics.

We apply modality-specific preprocessing pipelines following the configuration used in MultiGATE. For the RNA modality, if highly variable gene (HVG) annotations are already provided, they are used directly; otherwise, genes are first filtered and the top 3,000 HVGs are selected. When log-transformation is not present, total-count normalization followed by log1p transformation is applied, and the data are subsequently scaled. For the ATAC modality, we apply a standard preprocessing procedure with latent semantic indexing (LSI), retaining the top 50 LSI components as ATAC features. Finally, the latent representations of both modalities are z-score normalized prior to multimodal fusion.

B.4 Simulated Mouse Visual Cortex Dataset

The simulated mouse visual cortex dataset is constructed based on a spatially resolved multi-omics dataset from the mouse visual cortex, originally generated using STARmap spatial transcriptomics [Wang *et al.*, 2018]. The original data consist of 1,207 spatially resolved cells, each annotated with one of six cortical layer labels: L1, L2/3, L4, L5, L6, and HPC/CC, providing a well-defined layered tissue structure for benchmarking spatial domain identification. Building upon this dataset, COSMOS[Zhou *et al.*, 2025] introduced a simulation protocol to generate two complementary omics datasets, referred to as Omics-1 and Omics-2. In this process, cells from adjacent cortical layers (L4/L5 and L5/L6) are selectively shuffled in each simulated dataset, followed by the addition of Gaussian noise to gene expression values. As a

result, the two simulated omics datasets exhibit complementary but incomplete separability: cells from L4/L5 cannot be reliably distinguished using Omics-1 alone, while cells from L5/L6 cannot be separated using Omics-2 alone. This design creates a challenging multimodal integration scenario that explicitly tests a model’s ability to robustly integrate heterogeneous and noisy signals.

In this work, we use both the simulated data and the corresponding ground-truth annotations provided by COSMOS, where the original cortical layer labels are retained as evaluation references. For preprocessing, we apply the same pipeline to both modalities. Genes expressed in fewer than 10 cells are first filtered out to remove low-expression noise. If a variable-gene indicator is available, only the marked highly variable genes are retained. When raw counts are present without prior log-transformation, a log1p transformation is applied. Genes with near-zero variance are removed, and the remaining features are scaled at the gene level. Finally, any NaN or infinite values are replaced with finite values to ensure numerical stability during model training.

B.5 Spatial Simulated Triplet Omics Dataset

The spatial simulated triplet omics dataset is obtained from the benchmark provided by SpatialGlue [Long *et al.*, 2024], which is constructed using the nonnegative spatial factorization (NSF) protocol [Townes and Engelhardt, 2023]. This dataset is designed to evaluate multimodal integration under controlled settings and includes three complementary modalities: transcriptomics (RNA), chromatin accessibility (ATAC), and protein abundance. The dataset contains 1,296 spatial spots, with 1,000 genes in the RNA modality, 100 chromatin accessibility peaks in the ATAC modality, and 100 protein features, resulting in a balanced yet heterogeneous tri-modal integration scenario. Both the simulated data and the corresponding ground-truth spatial domain annotations are provided by the SpatialGlue study and are used directly in our experiments. The availability of three modalities makes this dataset particularly suitable for assessing the scalability of fusion methods beyond bi-modal settings, as well as their ability to jointly integrate molecular signals with distinct statistical properties.

For preprocessing, all three modalities are processed using consistent modality-specific pipelines following the configurations adopted in SpatialGlue. Each modality is normalized and standardized independently to obtain comparable latent representations. Specifically, transcriptomic and chromatin accessibility features are filtered and transformed to remove low-quality or low-variance signals, while protein features are standardized to account for differences in scale. After preprocessing, all modalities are z-score normalized before being used as inputs to multimodal fusion models.

C Evaluation Metrics

To quantitatively evaluate the quality of spatial domain identification, we employ six widely used **supervised clustering metrics** by comparing predicted cluster assignments with ground-truth (GT) annotations. These metrics jointly assess clustering consistency, information agreement, and robust-

Noise (σ)	ILR-SMO (Ours)						SpatialGlue					
	ARI(%)	Homo.(%)	NMI(%)	AMI(%)	V-meas.(%)	MI(%)	ARI(%)	Homo.(%)	NMI(%)	AMI(%)	V-meas.(%)	MI(%)
0.3	59.55	74.23	70.62	70.57	70.62	148.32	34.91	48.68	48.96	48.87	48.96	97.26
0.4	58.46	73.25	69.89	69.83	69.89	146.36	32.86	45.92	46.35	46.26	46.35	91.75
0.5	59.31	74.18	70.50	70.44	70.50	148.22	34.56	47.43	48.20	48.11	48.20	94.77
0.6	55.55	68.07	66.16	66.09	66.16	136.01	31.44	43.62	44.12	44.02	44.12	87.15
0.7	53.99	68.09	65.83	65.77	65.83	136.05	30.80	42.58	42.98	42.88	42.98	85.09
0.8	56.73	69.92	67.16	67.10	67.16	139.70	26.45	38.81	38.89	38.78	38.89	77.54
0.9	57.38	67.64	66.69	66.63	66.69	135.15	17.74	30.83	32.35	32.23	32.35	61.60
1.0	54.00	69.62	66.15	66.09	66.15	139.10	22.03	30.88	31.82	31.70	31.82	61.71
1.1	50.84	66.74	64.18	64.12	64.18	133.34	20.02	30.85	31.06	30.93	31.06	61.65
1.2	51.42	66.80	63.46	63.39	63.46	133.46	21.16	28.80	28.78	28.66	28.78	57.53
1.3	50.19	64.21	62.34	62.26	62.34	128.30	15.88	22.67	21.91	21.78	21.91	45.29
1.4	48.55	66.52	62.97	62.90	62.97	132.91	18.17	25.92	26.51	26.38	26.51	51.79
1.5	48.72	64.79	61.45	61.38	61.45	129.46	12.79	20.26	20.17	20.03	20.17	40.48
1.6	49.38	63.96	61.56	61.49	61.56	127.79	12.53	18.78	18.59	18.44	18.59	37.53

Table 1: Robustness analysis under different noise levels (σ).

ness to chance, and are commonly adopted in spatial transcriptomics and multi-omics studies.

Adjusted Rand Index (ARI). Adjusted Rand Index measures pairwise agreement between predicted clusters and GT labels while correcting for chance:

$$\text{ARI} = \frac{\sum_{ij} \binom{n_{ij}}{2} - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}{\frac{1}{2} \left[\sum_i \binom{a_i}{2} + \sum_j \binom{b_j}{2} \right] - \frac{\sum_i \binom{a_i}{2} \sum_j \binom{b_j}{2}}{\binom{n}{2}}}, \quad (43)$$

where n_{ij} is the number of samples shared by GT class i and predicted cluster j , a_i and b_j are marginal sums, and n is the total number of samples.

Homogeneity (Homo.). Homogeneity measures whether each predicted cluster contains samples from a single GT class. It is defined as

$$\text{Homo.} = 1 - \frac{H(C | K)}{H(C)}, \quad (44)$$

where C denotes the GT labels, K denotes the predicted clusters, $H(C | K)$ is the conditional entropy of C given K , and $H(C)$ is the entropy of the GT labels.

Mutual Information (MI). Mutual Information quantifies the shared information between GT labels and predicted clusters:

$$\text{MI}(C, K) = \sum_{c \in C} \sum_{k \in K} p(c, k) \log \frac{p(c, k)}{p(c) p(k)}, \quad (45)$$

where $p(c, k)$ denotes the joint probability of class c and cluster k .

V-measure (V-meas.). V-measure is the harmonic mean of homogeneity and completeness:

$$\text{V-meas.} = \frac{2 \cdot \text{Homo.} \cdot \text{Comp.}}{\text{Homo.} + \text{Comp.}}, \quad (46)$$

where completeness evaluates whether all samples of a GT class are assigned to the same cluster.

Adjusted Mutual Information (AMI). Adjusted Mutual Information corrects MI for chance agreement:

$$\text{AMI}(C, K) = \frac{\text{MI}(C, K) - \mathbb{E}[\text{MI}(C, K)]}{\max\{H(C), H(K)\} - \mathbb{E}[\text{MI}(C, K)]}, \quad (47)$$

where $\mathbb{E}[\text{MI}(C, K)]$ denotes the expected MI under a random labeling model.

Normalized Mutual Information (NMI). Normalized Mutual Information scales MI by the average entropy of the two labelings:

$$\text{NMI}(C, K) = \frac{2 \cdot \text{MI}(C, K)}{H(C) + H(K)}. \quad (48)$$

These metrics collectively provide a comprehensive assessment of clustering quality from complementary perspectives, enabling robust comparison of spatial multi-omics integration methods.

D Implementation Details

D.1 Model Training

ILR-SMO follows a multi-stage training procedure. In Stage I (Modality-Specific Pretraining), each modality is independently encoded using a dedicated autoencoder, where molecular modalities (e.g., RNA, ATAC, protein) are processed by multilayer perceptron-based encoders and histological images are processed by an image autoencoder when applicable. At this stage, each encoder is optimized solely with a reconstruction loss to obtain modality-specific latent features that preserve essential information.

In Stage II (Iterative Latent Refinement and Fusion), pretrained modality-specific latent features are integrated through the iterative latent refinement mechanism. Multi-modal fusion is performed via Sequential Latent Updates (SLU), where modality information is injected into a shared latent representation in a progressive manner. To account for uneven modality quality, reliability-aware modality gating is applied to adaptively modulate modality contributions during refinement.

Order	ARI(%)	Homo.(%)	MI(%)	V-meas.(%)	AMI(%)	NMI(%)	Epochs
RNA–ATAC–Protein	100.00	100.00	154.74	100.00	100.00	100.00	25
RNA–Protein–ATAC	100.00	100.00	154.74	100.00	100.00	100.00	15
ATAC–RNA–Protein	100.00	100.00	154.74	100.00	100.00	100.00	35
ATAC–Protein–RNA	100.00	100.00	154.74	100.00	100.00	100.00	15
Protein–RNA–ATAC	100.00	100.00	154.74	100.00	100.00	100.00	15
Protein–ATAC–RNA	100.00	100.00	154.74	100.00	100.00	100.00	25

Table 2: Effect of modality update order in Sequential Latent Updates (SLU) on the spatial simulated triplet-omics dataset. All permutations achieve identical clustering performance; differences are observed only in convergence speed.

The overall training objective is formulated as stability-aware joint optimization, which combines multiple loss components, including reconstruction loss, cross-modality alignment loss, gating-weighted alignment loss, Spatial Consistency Regularization based on k-nearest-neighbor spatial relationships, representation dispersion regularization to prevent representation collapse, and a gating entropy regularizer to avoid degenerate modality weighting. The relative weights of these loss terms are specified per dataset.

All models are trained using the AdamW optimizer with gradient clipping applied to stabilize optimization. During training, model checkpoints are evaluated periodically, and the best-performing model is selected based on clustering performance.

D.2 Clustering and Evaluation

After training, the refined shared latent representation is used for spatial domain identification. Clustering is performed using either KMeans or Leiden clustering, depending on the dataset, and clustering quality is evaluated using Adjusted Rand Index (ARI) and other metrics reported in the main text. The model checkpoint achieving the best validation performance is retained for final evaluation.

D.3 Data Preprocessing and Input Construction

All datasets are processed using a unified preprocessing pipeline. Molecular modalities are normalized and transformed using modality-specific procedures, including gene filtering, optional highly variable gene selection, log-transformation, scaling, or latent semantic indexing for chromatin accessibility features. Protein features are standardized to account for scale differences. For datasets with histological images, dual-scale image patches are extracted from tissue sections based on spatial coordinates and encoded using pretrained vision models to obtain fixed-length embeddings.

After preprocessing, features from each modality are organized into a dictionary and converted into tensors of shape $(N, 1, D)$, where N denotes the number of spatial locations and D the modality-specific feature dimension. Spatial coordinates and ground-truth annotations, when available, are stored alongside the input features and used for spatial regularization and evaluation.

E Detailed Results under Gaussian Noise Perturbation

This section reports detailed quantitative results for the robustness experiment with Gaussian noise perturbation, complementing the ARI-based comparison shown in the main text. Following the setup in Section 4.4, Gaussian noise with standard deviation σ from 0.3 to 1.6 is injected into the ATAC modality of the mouse brain dataset, while all other modalities and training configurations remain unchanged.

Tables 1 summarize the clustering performance of SpatialGlue and ILR-SMO, respectively, evaluated using six standard metrics: ARI, NMI, AMI, V-measure, Homogeneity, and Mutual Information. While the main text highlights ARI for clarity, these results show that ILR-SMO consistently maintains higher and more stable performance across all metrics as noise intensity increases, while SpatialGlue exhibits substantial degradation under stronger perturbations.

F Sensitivity to Modality Update Order

This experiment examines whether the performance of Sequential Latent Updates (SLU) depends on the order in which modalities are integrated. We evaluate all possible permutations of modality update order on the spatial simulated triplet-omics dataset, which contains RNA, ATAC, and protein modalities, resulting in six distinct orders. All experiments use identical configurations except for the modality order.

As shown in Table 2, all modality orders achieve identical clustering performance, reaching 100% ARI and identical scores across all evaluation metrics. The only observable difference lies in convergence speed during Stage II optimization, where the number of epochs required in stage II to reach convergence ranges from 15 to 35 across different orders. These results indicate that ILR-SMO is insensitive to modality update order, and that SLU does not rely on a specific ordering to achieve optimal solutions.

G Model Complexity and Computational Cost

We compare the parameter count and computational cost of ILR-SMO with representative spatial multi-omics integration methods on the Human Lymph Node dataset. Computational

Method	Params (M)	FLOPs (G)
MultiVI	0.38	2.77
SpatialGlue	0.02	0.18
MISO	0.04	0.13
PRAGA	0.01	84.62
COSMOS	0.16	0.55
SWITCH	1.21	6.79
MultiGATE	9.29	66.04
ILR-SMO (Ours)	14.10	52.03

Table 3: Model complexity comparison on the Human Lymph Node dataset. FLOPs are computed under identical experimental settings.

cost is measured in floating-point operations (FLOPs) under identical input resolutions and batch settings.

As summarized in Table 3, ILR-SMO contains 14.10M parameters and requires 52G FLOPs, which is modest in scale compared to contemporary deep learning models commonly used in computer vision or multimodal learning. Although ILR-SMO adopts a modular architecture and has a relatively larger parameter count than some baselines, its computational cost remains comparable to or lower than several recent spatial multi-omics methods. In practice, all experiments can be efficiently conducted on a single NVIDIA GeForce RTX 3090 GPU, indicating that iterative latent refinement does not introduce prohibitive computational overhead and remains accessible under standard research hardware settings.

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